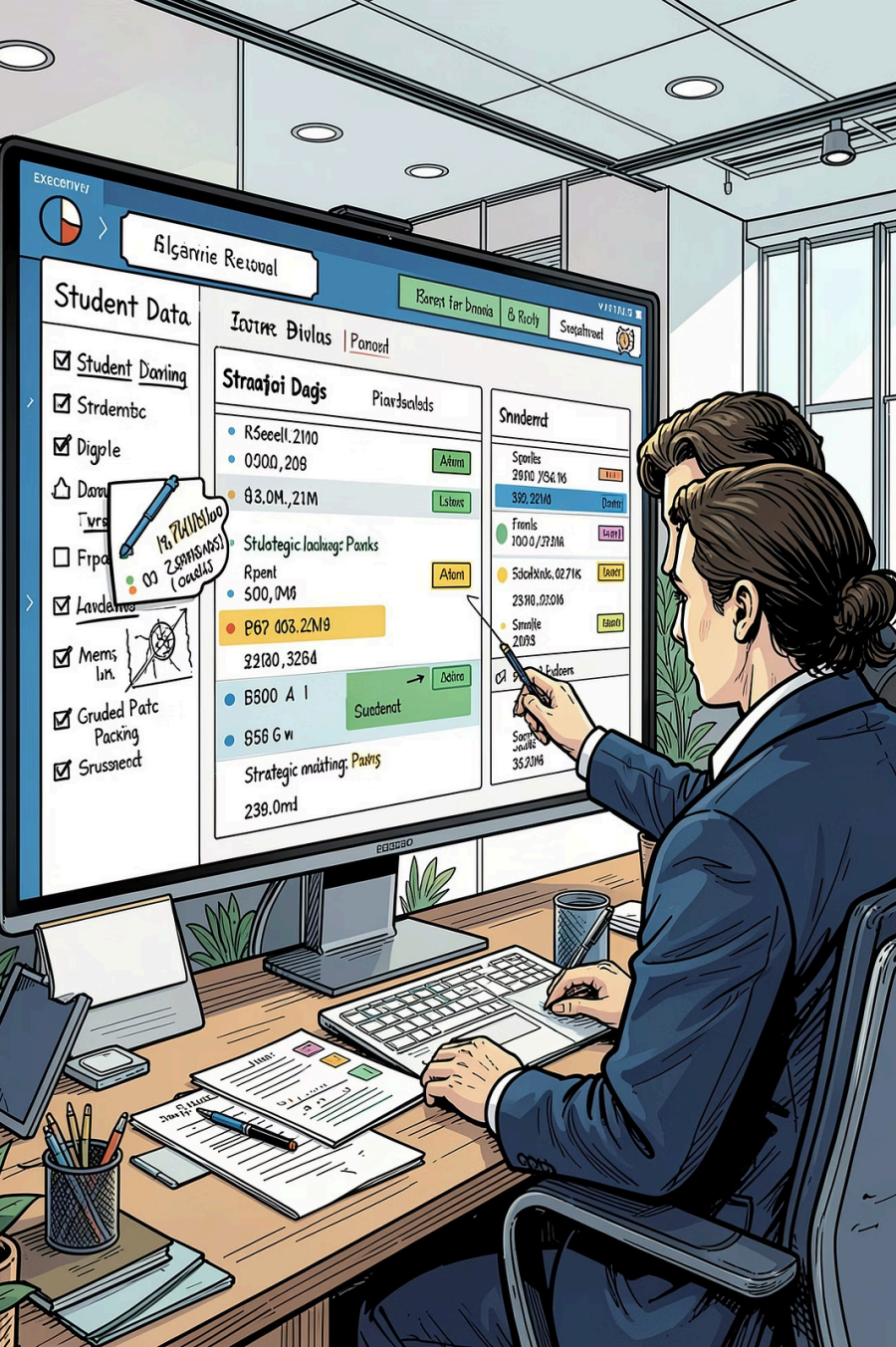




EdTech Student Engagement & Churn Mitigation

Product Analytics Report & PRD · 2,800 Active Students · Growth & Retention Track



Executive Summary

Core objective: isolate, diagnose, and remediate student churn vectors.
Three risk tiers identified across 2,800 active students — with a projected **\$392,305 in reclaimable annual ARR** at scale.

1,360

Low Risk

48.57% of cohort

847

Medium Risk

30.25% of cohort

593

High Risk

21.18% of cohort

Platform Baseline Metrics

Behavioral and performance averages across the full 2,800-student cohort.

7.50

Weekly Logins

Sessions per week average

70.13%

Quiz Score

Average performance

50.60%

Course Progress

Average completion ratio

5.61

Assignments

Items per student

60.26%

Video Watch Ratio

Average video consumption

The Churn Paradox

High-risk students are **not** failing academically — they have higher progress (53.20% vs. 49.97%) and quiz scores (71.01% vs. 69.70%) than low-risk peers. The real driver: **social isolation and cognitive burnout**.

Course Progress Correlation

+0.0373 — nearly zero link to churn

Quiz Score Correlation

+0.0289 — academic success doesn't prevent attrition

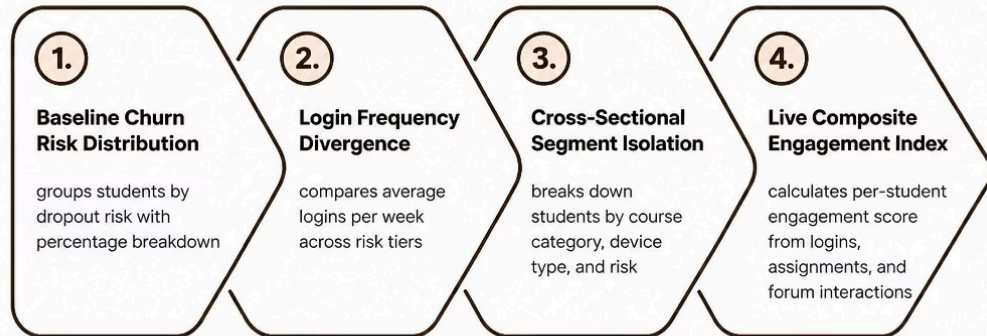
Forum Interactions

-0.0184 — strongest negative correlation; community engagement is the key retention lever



SQL Metrics Pipeline

Four automated queries from EdTech Learning Platform Analytics.sql power operational dashboards.



Query Purposes

- **Baseline Distribution:** Churn risk split with percentages
- **Login Divergence:** Engagement velocity by risk tier
- **Segment Isolation:** Risk by course category × device
- **Engagement Index:** Composite score per student



Power BI Dashboard Design

Page 1: Platform Overview

2.8K students · 70.13% quiz baseline · 50.60% progress.
Design (581) and AI & ML (567) lead; Business (528) is leanest.

Page 2: Dropout Risk Analysis

High-risk: 7.4 visits/week, 3.4K assignments, 35K video units — vs. 7.7K assignments and 82K units for low-risk.

Page 3: Course & Device Matrix

Near-even device split: Desktop 957, Tablet 924, Mobile 919. Highest risk concentrations: Design (139) and Data Science (133).

Intervention: "Co-Learn" Micro-Communities



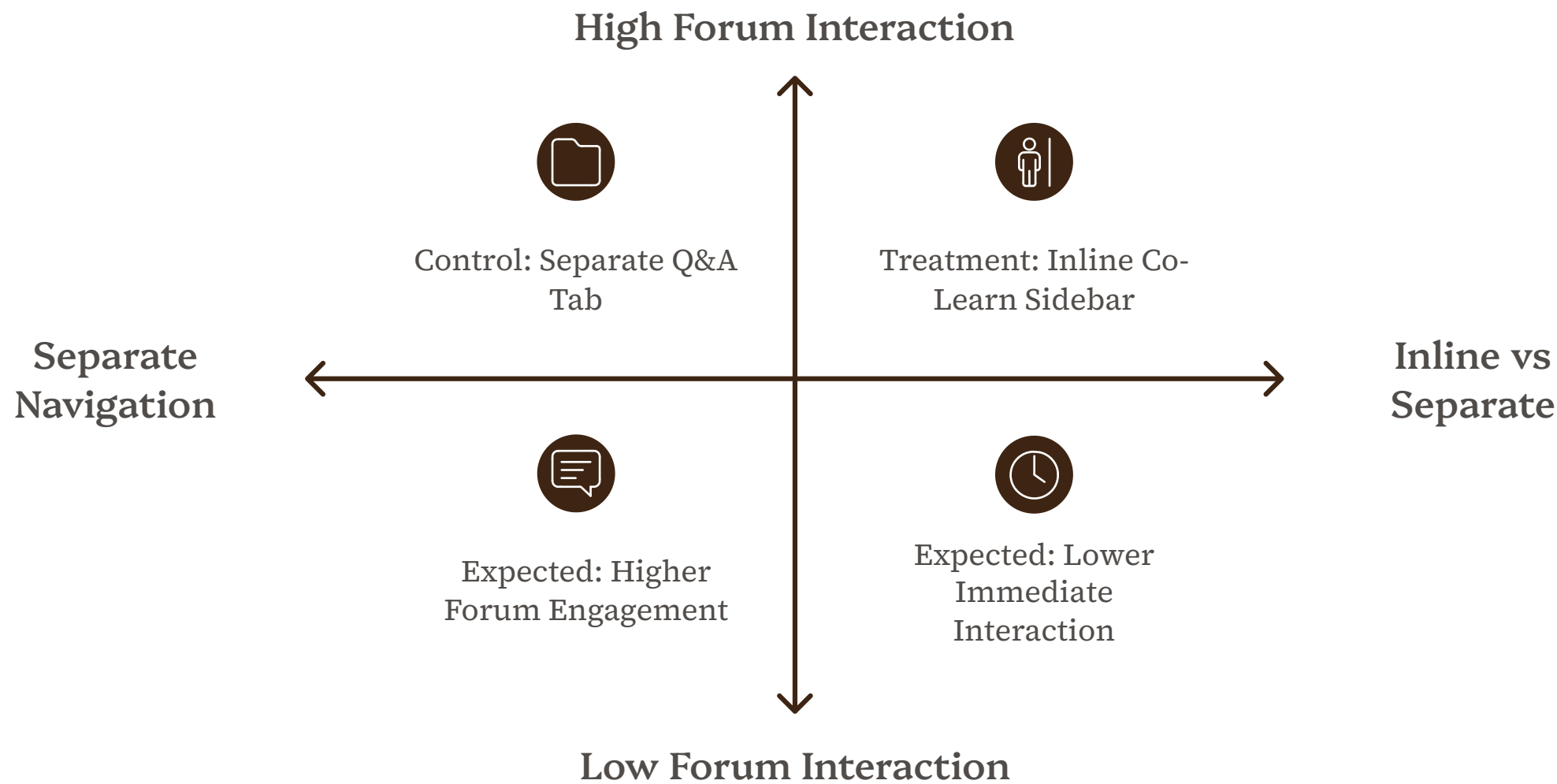
The Problem

High-risk users progress in silos, skip forums, and abandon the platform at friction points.

The Solution

When a student enters a technical track (e.g., Data Science or Design), the platform auto-spins a contextual sidebar connecting them to **15–20 peers** on the same module step — inline, without leaving the learning flow.

A/B Testing Framework



Control (Branch A) uses the standard layout requiring navigation to a separate tab; Treatment (Branch B) adds the inline Co-Learn sidebar next to the video player.

Core Hypothesis (H₁)

Inline sidebar increases forum interactions by $\geq 25\%$, reducing 30-day high-risk dropout by $\geq 5\%$

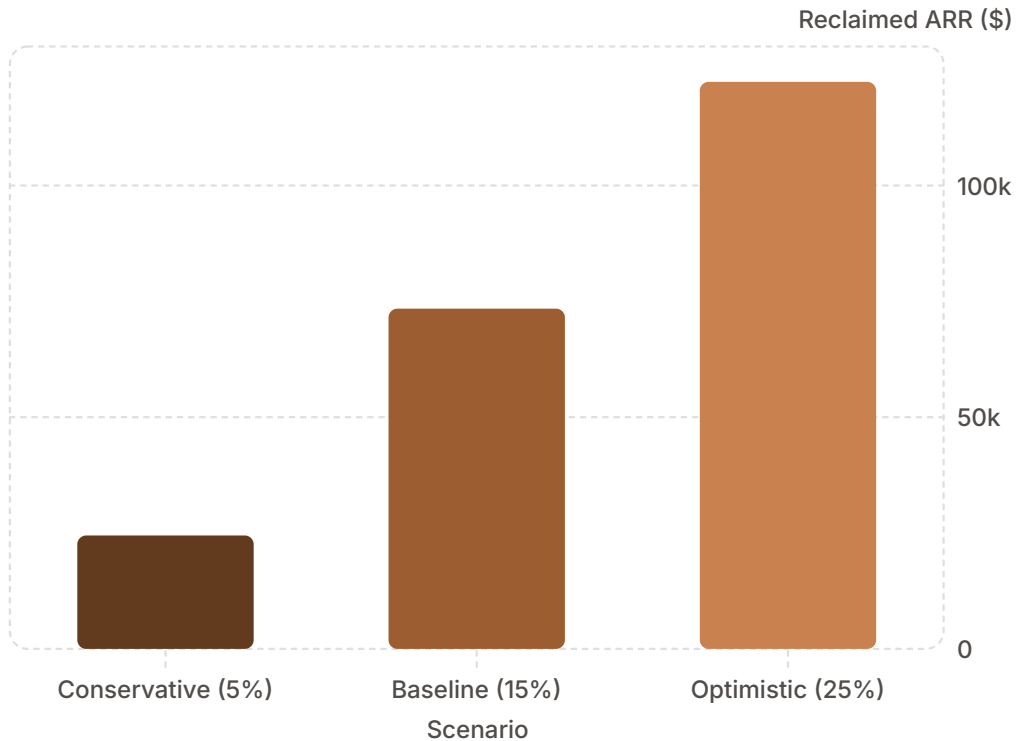
Sample Size

1,100 students per branch · 50/50 split · 80% power, 95% confidence ($\alpha = 0.05$)

Guardrail Metric

Video Watch Ratio must stay $\geq 60.26\%$ — sidebar must not degrade playback

Business Case & Financial Projections



ARR Risk Exposure

593 high-risk students $\times$ 75% expected churn $\times$ **\$1,100/seat** = **\$489,500** in vulnerable ARR.

Strategic Scale Impact

Scaled to Edmentum's ~15,000 institutional seats, the **15% baseline case** yields **\$392,305** in reclaimable annual revenue — a high-priority, high-ROI growth project.

Functional Specifications (PRD)

Feature Requirements

→ Automated Grouping Engine

Auto-sort users into 15–20 peer clusters on target course registration

→ Inline Chat UI

Collapsible sidebar integrated directly into the video player layout

→ Contextual Threading

Discussions filtered to the user's active module step only

Performance SLOs

- **Message Load Latency:** $\leq 150\text{ms}$ for real-time stream
- **Thread Decoupling:** Sidebar loads independently from video thread — chat activity never interrupts playback or degrades video consumption standards